

CSC435 Programming Assignment - I

1. Linux Ecosystem and Virtual Machines

A. Commands for retrieving system info.

- lscpu (For getting CPU Info)
- free -h (For Getting total amount of RAM)
- df -h (For Getting total amount of Disk Storage)

Screen Snips for the above commands

```
kedar_vm@kedar-server:~$ free -h
```

	total	used	free	shared	buff/cache	available
Mem:	1.9Gi	263Mi	1.1Gi	1.0Mi	590Mi	1.5Gi
Swap:	2.0Gi	0B	2.0Gi			

```
kedar_vm@kedar-server:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	195M	1.1M	194M	1%	/run
/dev/mapper/ubuntu--vg-ubuntu--lv	23G	6.2G	16G	29%	/
tmpfs	974M	0	974M	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/sda2	2.0G	130M	1.7G	8%	/boot
/dev/sda1	1.1G	6.1M	1.1G	1%	/boot/efi
tmpfs	195M	4.0K	195M	1%	/run/user/1000

```
kedar_vm@kedar-server:~$ lscpu
```

Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 39 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: 13th Gen Intel(R) Core(TM) i9-13900H
CPU family: 6
Model: 186
Thread(s) per core: 1
Core(s) per socket: 2
Socket(s): 1
Stepping: 2
BogoMIPS: 5990.39
Flags: fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse ss e2 ht syscall nx rdtscp lm constant_tsc rep_good nopl xtopology nonstop_tsc cpuid tsc_known_freq pni pclmulqdq sse3 cx16 sse4_1 sse4_2 movbe popcnt aes rdrand hypervisorlahf_lm abm 3dnowpr capabilities efetch ibrs_enhanced fsgsbase bmi1 bmi2 invpcid rdseed clflushopt arat md_clear flush_lld arch_capabilities

Virtualization features:
Hypervisor vendor: KVM
Virtualization type: full

Caches (sum of all):
L1d: 64 KiB (2 instances)
L1i: 128 KiB (2 instances)
L2: 4 MiB (2 instances)
L3: 48 MiB (2 instances)

NUMA:
NUMA node(s): 1
NUMA node0 CPU(s): 0,1

Vulnerabilities:
Gather data sampling: Not affected
Itlb multihit: Not affected
L1tf: Not affected
Mds: Not affected
Meltdown: Not affected
Mmio stale data: Not affected
Retbleed: Mitigation; Enhanced IBRS
Spec rstack overflow: Not affected
Spec store bypass: Vulnerable
Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization
Spectre v2: Mitigation; Enhanced IBRS, RSB filling, PBRSE-IBRS SW sequence
Srbds: Not affected
Tsx async abort: Not affected

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B. Commands used for enabling SSH access to the Virtual Machine.

- Before installing anything first run “sudo apt update” command.
- Now run “sudo apt install openssl” command to install openssl in Ubuntu server (virtual machine).
- Then type “sudo service ssh start” command to start running ssh service.
- To check the ssh status run “sudo service ssh status” command.
- Optionally to start ssh on boot, “sudo systemctl enable ssh” command can be used.

Screen Snips for the above commands.

```
kedar_vm@kedar-server:~$ sudo apt update
[sudo] password for kedar_vm:
Hit:1 http://in.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://in.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://in.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://in.archive.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
52 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

```
kedar_vm@kedar-server:~$ sudo apt install openssl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
openssl is already the newest version (3.0.2-0ubuntu1.12).
openssl set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 52 not upgraded.
```

```
kedar_vm@kedar-server:~$ sudo service ssh start
kedar_vm@kedar-server:~$ sudo service ssh status
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2024-01-24 06:51:13 UTC; 1h 43min ago
```

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```
kedar_vm@kedar-server:~$ sudo systemctl enable ssh
Synchronizing state of ssh.service with SysV service script with /lib/systemd/systemd-sysv-install.
Executing: /lib/systemd/systemd-sysv-install enable ssh
```

- You can try running the command – “ssh your_username@your_server_ip”
- C. Commands used to turn on the firewall and to open the SSH port on the firewall.
- Before installing anything first run “sudo apt update” command.
 - Now run “sudo apt install ufw” command to install Uncomplicated Firewall (ufw) in Ubuntu server (virtual machine).
 - After installation run “sudo ufw enable” to Enable firewall.
 - Use “sudo ufw allow “number”” to allow (the specific) port.
 - Use “sudo ufw deny “number”” to remove/deny (the specific) port.
 - Use “sudo ufw status to check the firewall and port status.

Screen Snips for the above commands.

```
kedar_vm@kedar-server:~$ sudo apt install ufw
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
ufw is already the newest version (0.36.1-4ubuntu0.1).
ufw set to manually installed.
0 upgraded, 0 newly installed, 0 to remove and 52 not upgraded.
```

```
kedar_vm@kedar-server:~$ sudo ufw enable
[sudo] password for kedar_vm:
Firewall is active and enabled on system startup
```

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```
kedar_vm@kedar-server:~$ sudo ufw enable
Firewall is active and enabled on system startup
kedar_vm@kedar-server:~$ sudo ufw allow 22
Rule added
Rule added (v6)
kedar_vm@kedar-server:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
kedar_vm@kedar-server:~$ sudo ufw allow out 80
Rule added
Rule added (v6)
kedar_vm@kedar-server:~$ sudo ufw status
Status: active

To Action From
--
22 ALLOW Anywhere
22/tcp ALLOW Anywhere
22 (v6) ALLOW Anywhere (v6)
22/tcp (v6) ALLOW Anywhere (v6)

80 ALLOW OUT Anywhere
80 (v6) ALLOW OUT Anywhere (v6)

kedar_vm@kedar-server:~$ _
```

- D. Commands used to retrieve IP address of the virtual machine.
- **IP (Internet Protocol) address:** An IP address is a numerical label assigned to each device participating in a computer network that uses the Internet Protocol for communication. It serves two main purposes: host or network interface identification and location addressing.
 - **TCP/UDP port:** In computer networking, a port is a communication endpoint that identifies a specific process or service on a host device in a network. TCP (Transmission Control Protocol) and UDP (User Datagram Protocol) are transport layer protocols that use port numbers to identify specific applications or services on a host. Ports allow multiple applications to run on the same device without interference.
 - The default port for SSH (Secure Shell) is port 22.
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- Use “ip addr show” to show ip address(visible under enp0s8).

Screen Snips for the above commands.

```
kedar_vm@kedar-server:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c1:14:20 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 69244sec preferred_lft 69244sec
    inet6 fe80::a00:27ff:fec1:1420/64 scope link
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:10:9d:a4 brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.1/24 brd 192.168.56.255 scope global enp0s8
        valid_lft forever preferred_lft forever
    inet6 fe80::a00:27ff:fe10:9da4/64 scope link
        valid_lft forever preferred_lft forever
```

E. Commands for generating public and private keys.

- Use “ssh-keygen” to generate public and private keys (RSA algorithm).

Screen Snips for the above commands

```
kedar_wsl@FreakyS-PC:/$ ls /home/
kedar_wsl
kedar_wsl@FreakyS-PC:/$ ls /home/kedar_wsl
kedar_wsl@FreakyS-PC:/$ ls /home/kedar_wsl/.ssh/
id_rsa id_rsa.pub known_hosts known_hosts.old
kedar_wsl@FreakyS-PC:/$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/kedar_wsl/.ssh/id_rsa): /home/kedar_wsl/.ssh/id_temp_rsa
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kedar_wsl/.ssh/id_temp_rsa
Your public key has been saved in /home/kedar_wsl/.ssh/id_temp_rsa.pub
The key fingerprint is:
SHA256:4VVZkzihty9iK0ddhWivQ5G9QPBjX048B+5YEEhGiw0 kedar_wsl@FreakyS-PC
The key's randomart image is:
+----[RSA 3072]-----+
|      Eo==*X+o. |
|      =.*@++o  |
|      o =.=+ . |
|      . o +.Bo. |
|      S +.+B . |
|      o oo= o  |
|      . o ...  |
|      o        |
+----[SHA256]-----+
kedar_wsl@FreakyS-PC:/$ ls /home/kedar_wsl/.ssh/
id_rsa id_rsa.pub id_temp_rsa id_temp_rsa.pub known_hosts known_hosts.old
```

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- F. Commands used to enable password less authentication.
- First create a private and public keys in the localhost machine(wsl) using the command “ssh-keygen”.
 - Before performing the public id copy, make sure to change the sshd_config file present in /etc/ssh/sshd_config. (uncomment some sections of lines and switch yes/no for some configurations.)
 - Also add the virtual machine ip address in the network configuration file under the /etc/netplan/ folder, under enp0s8 line.
 - Now Copy the public key from local host(wsl) to host machine(virtual machine) using the command “ssh-copy-id <username>@<vm_ip>”.
 - Finally type the command “ssh <username>@<vm_ip>” and connect to the host system.

Screen Snips for the above commands.

```
kedar_wsl@FreakyS-PC:/$ ls /home/
kedar_wsl
kedar_wsl@FreakyS-PC:/$ ls /home/kedar_wsl
kedar_wsl@FreakyS-PC:/$ ls /home/kedar_wsl/.ssh/
id_rsa id_rsa.pub known_hosts known_hosts.old
kedar_wsl@FreakyS-PC:/$ ssh-keygen
Generating public/private rsa key pair.
Enter file in which to save the key (/home/kedar_wsl/.ssh/id_rsa): /home/kedar_wsl/.ssh/id_temp_rsa
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/kedar_wsl/.ssh/id_temp_rsa
Your public key has been saved in /home/kedar_wsl/.ssh/id_temp_rsa.pub
The key fingerprint is:
SHA256:4VVZkzihTy9ik0ddhWivQ5G9QPBjX048B+5YEEhGiw0 kedar_wsl@FreakyS-PC
The key's randomart image is:
+---[RSA 3072]-----+
|      Eo==*X+o. |
|      =.*@++o  |
|      o =.==+.  |
|      . o +.Bo.  |
|      S +.+B .  |
|      o oo= o   |
|      . o ...   |
|      o         |
+---[SHA256]-----+
kedar_wsl@FreakyS-PC:/$ ls /home/kedar_wsl/.ssh/
id_rsa id_rsa.pub id_temp_rsa id_temp_rsa.pub known_hosts known_hosts.old
```

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```
# This is the network config written by 'subiquity'
network:
  ethernets:
    enp0s3:
      dhcp4: true
    enp0s8:
      dhcp4: false
      addresses: [192.168.56.1/24]
  version: 2
```

```
# This is the sshd server system-wide configuration file. See
# sshd_config(5) for more information.

# This sshd was compiled with PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented. Uncommented options override the
# default value.

Include /etc/ssh/sshd_config.d/*.conf

#Port 22
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key

# Ciphers and keying
#RekeyLimit default none

# Logging
#SyslogFacility AUTH
#LogLevel INFO

# Authentication:

#LoginGraceTime 2m
#PermitRootLogin prohibit-password
#StrictModes yes
#MaxAuthTries 6
#MaxSessions 10

PubkeyAuthentication yes

# Expect .ssh/authorized_keys2 to be disregarded by default in future.
AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none
```

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```
# Expect .ssh/authorized_keys2 to be disregarded by default in future.
AuthorizedKeysFile .ssh/authorized_keys .ssh/authorized_keys2

#AuthorizedPrincipalsFile none

#AuthorizedKeysCommand none
#AuthorizedKeysCommandUser nobody

# For this to work you will also need host keys in /etc/ssh/ssh_known_hosts
#HostbasedAuthentication no
# Change to yes if you don't trust ~/.ssh/known_hosts for
# HostbasedAuthentication
#IgnoreUserKnownHosts no
# Don't read the user's ~/.rhosts and ~/.shosts files
#IgnoreRhosts yes

# To disable tunneled clear text passwords, change to no here!
PasswordAuthentication no
#PermitEmptyPasswords no

# Change to yes to enable challenge-response passwords (beware issues with
# some PAM modules and threads)
KbdInteractiveAuthentication no

# Kerberos options
#KerberosAuthentication no
#KerberosOrLocalPasswd yes
#KerberosTicketCleanup yes
#KerberosGetAFSToken no

# GSSAPI options
#GSSAPIAuthentication no
#GSSAPICleanupCredentials yes
#GSSAPIStrictAcceptorCheck yes
#GSSAPIKeyExchange no

# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin without-password".
```

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```
# Set this to 'yes' to enable PAM authentication, account processing,
# and session processing. If this is enabled, PAM authentication will
# be allowed through the KbdInteractiveAuthentication and
# PasswordAuthentication. Depending on your PAM configuration,
# PAM authentication via KbdInteractiveAuthentication may bypass
# the setting of "PermitRootLogin without-password".
# If you just want the PAM account and session checks to run without
# PAM authentication, then enable this but set PasswordAuthentication
# and KbdInteractiveAuthentication to 'no'.
UsePAM yes

#AllowAgentForwarding yes
#AllowTcpForwarding yes
#GatewayPorts no
X11Forwarding yes
#X11DisplayOffset 10
#X11UseLocalhost yes
#PermitTTY yes
PrintMotd no
#PrintLastLog yes
#TCPKeepAlive yes
#PermitUserEnvironment no
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem sftp /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
```

```
kedar_wsl@FreakyS-PC:/$ ssh-copy-id kedar_vm@192.168.56.1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/kedar_wsl/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed

/usr/bin/ssh-copy-id: WARNING: All keys were skipped because they already exist on the remote system.
(if you think this is a mistake, you may want to use -f option)
```

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```
kedar_wsl@FreakyS-PC:~$ ssh kedar_vm@192.168.56.1
Welcome to Ubuntu 22.04.3 LTS (GNU/Linux 5.15.0-91-generic x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:       https://ubuntu.com/advantage

System information as of Wed Jan 24 12:39:19 PM UTC 2024

System load:  0.0                Processes:            120
Usage of /:   26.7% of 22.94GB   Users logged in:     0
Memory usage: 12%               IPv4 address for enp0s3: 10.0.2.15
Swap usage:   0%                IPv4 address for enp0s8: 192.168.56.1

Expanded Security Maintenance for Applications is not enabled.

45 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Wed Jan 24 08:23:56 2024
kedar_vm@kedar-server:~$ |
```

2. Clean Dataset

A.

- **Wall Time (Real Time):**
- Wall time refers to the actual elapsed time that a program takes to execute from start to finish, including all the time spent waiting for resources, such as I/O (input/output) operations or time spent waiting in a queue for CPU time.
- It is the total time a user would perceive, including all delays and waits.
- Wall time is the most intuitive measure of how long a program takes to run in real-world terms.

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- **CPU Time (Processor Time):**
- CPU time, on the other hand, measures the amount of time the CPU spends actively executing a program's instructions.
- It excludes time spent waiting for I/O operations or other resources.
- It represents the actual computational work done by the program
- **Method Used:**
- `std::chrono::high_resolution_clock::now();`

B.

- **Dataset1**
- **MB:** 192.889
- **MiB:** 183.954

- **Dataset5**
- **MB:** 1704.38
- **MiB:** 1625.43

C.

- **System Specifications:**
- **Type** - SSD
- **Read speed** - 32.8 KB/s
- **Write speed** - 722 KB/s
- **Active time** - 5%
- **Average response time** - 14.4 ms
- Disk speed directly influences the program as half of the program is reading and writing to a file and the operation takes place till the end of the program.

D.

- **Command Used:**
- `sudo sync`
- `sudo sh -c 'echo 3 > /proc/sys/vm/drop_caches'`

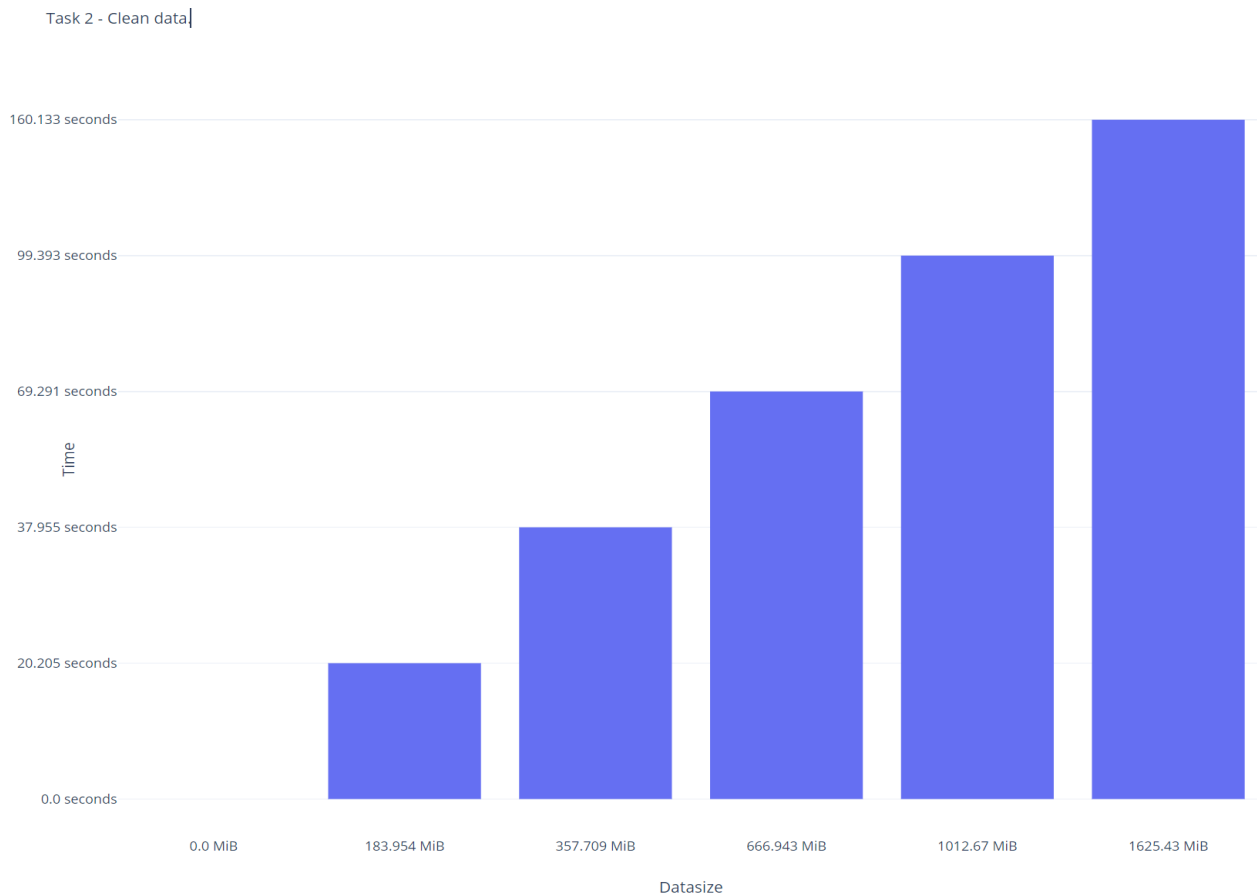
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- The dataset size can significantly influence the performance of a program on a virtual machine (VM) due to several factors related to memory, storage, and processing power.

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E. Task 2 Graph



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3.

Count Dataset

A.

- **Datastructure:**
- I have use `std::unordered_map`.
- It helps in finding the elements quickly and it stores elements as key and value pairs.
- Unordered map doesn't store elements in any order as it is optimized for quick look-up(s).

B.

- **Compute-Intensive Applications:**
- **Characteristics:** These applications primarily rely on the processing power of the CPU (Central Processing Unit).
- **Usage:** Tasks that involve complex calculations, simulations, mathematical modeling, and other operations that require a significant amount of computational power.
- **Example:** Scientific simulations, rendering in computer graphics, financial modeling, and some artificial intelligence/machine learning computations.
- **Memory-Intensive Applications:**
- **Characteristics:** These applications require a substantial amount of system memory (RAM) to perform efficiently.
- **Usage:** Handling large datasets, working with high-resolution images or videos, database applications, and applications that involve frequent read and write operations to memory.

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- **Example:** Database servers, in-memory caching systems, and image/video editing software.
- **I/O-Intensive Applications:**
- **Characteristics:** These applications rely heavily on Input/Output operations, involving data transfers between the system and external storage devices or networks.
- **Usage:** Applications that involve frequent reading and writing of data to storage, network communication, and other I/O operations. The efficiency of data transfer becomes crucial for performance.
- **Example:** File servers, web servers, streaming services, and applications that deal with large-scale data input/output.

C.

- **Stats of my program:**
- My program is actually **Compute-Intensive, Memory-Intensive, I/O-Intensive** depending what stage the program is at.
- There are 4 stages of the program.
- First stage is loading the contents of a text file into a buffer. In this stage my program is **I/O-Intensive**.
- Second stage is counting the elements, this is **Compute-Intensive** as CPU needs to do counting based on the file text data.
- Third stage is appending all the data (as we prepare to do a file push). This is **Memory-Intensive** as all the words and the word repetition count numbers needs to be appended to a single string.
- Fourth stage is again **Memory-Intensive** as we are pushing the string output data into the file.

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D.

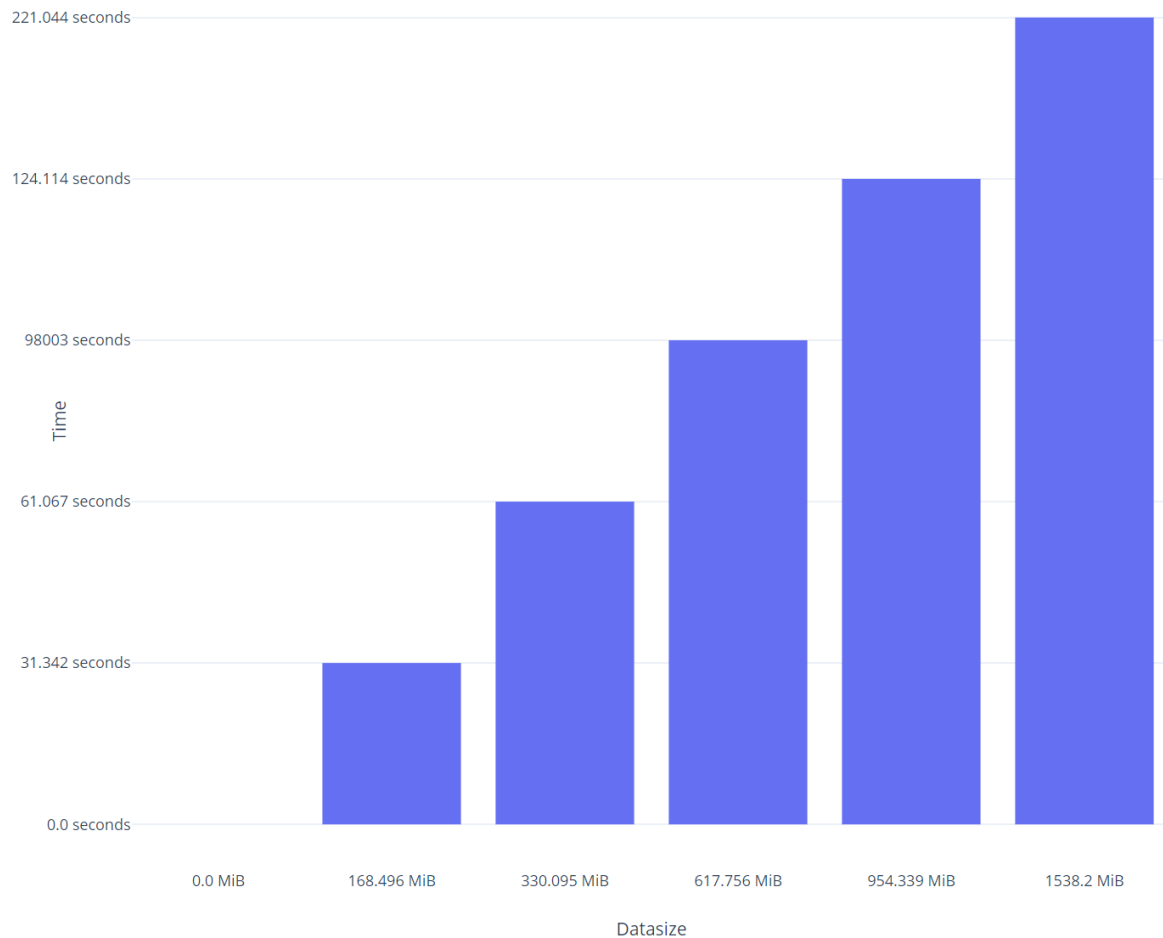
- The dataset size can significantly influence the performance of a program on a virtual machine (VM) due to several factors related to memory, storage, and processing power.

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E. Task 3 Graph

Task 3 - Count data.



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4.

Sort Dataset

A.

- **Datastructure:**
- I have use std:: map.
- It helps in finding the elements quickly and it stores elements as key and value pairs.
- Map stores elements in any user provided order.

B.

- **Algorithm:**
- As I have used std:: map, I have the option to insert the elements in a sorted order.
- The Syntax which can be used to store the element counter in descending order is as follows: `std::map<int, std::string, std::greater<int>>& wordMap`.
- As I have used the `std::map<int, std::string, std::greater<int>>& wordMap` syntax, when I insert a pair, the element is inserted at the correct sorted position.

C.

- **Stats of my program:**
- My program is actually **Compute-Intensive, Memory-Intensive, I/O-Intensive** depending what stage the program is at.
- There are 4 stages of the program.
- First stage is loading the contents of a text file into a buffer. In this stage my program is **I/O-Intensive**.
- Second stage is populating and sorting the elements into the map, this is **Compute-Intensive and Memory-Intensive** as CPU needs to do sort and insert the data based on the file text data.

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- Third stage is appending all the data (as we prepare to do a file push). This is **Memory-Intensive** as all the words and the word repetition count numbers needs to be appended to a single string.
- Fourth stage is again **Memory-Intensive** as we are pushing the string output data into the file.

D.

- The total number of words can influence the performance of a program on a virtual machine (VM), because as number of elements increase more amount of work needs to be done.

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E. Task 4 Graph

Task 4 - Sort data.

